

NumPy Functions and Operations - Examples

NumPy Functions	Description
<code>np.array([1,2,3])</code>	Creates a 1D array of shape 1x3 with values 1, 2, 3
<code>np.array([(1,2,3), (4,5,6)])</code>	Creates 2D array of shape 2x3 with values 1,2,3,4,5,6
<code>np.array([[(1,2,3), (4,5,6)], [(7,8,9), (10,11,12)]])</code>	Creates a 3D array with shape 2x2x3
<code>np.zeros(3,4)</code>	Creates a 3x4 array of zeros
<code>np.arange(1,60,5)</code>	Creates a 1D array of values 1 through 60 at steps of 5
<code>arr.reshape(2,3,4)</code>	Reshapes array arr into an array of shape 2x3x4
<code>arr.shape</code>	To get the shape of the array arr

NumPy Operations	Description
<code>a[0]</code>	Gets the 0th element in 1D array
<code>b[0,0]</code>	Gets the 0th element in 2D array
<code>c[0,0,0]</code>	Gets the 0th element in 3D array
<code>c[9,0,0]</code>	Gets the element which is the 0th element in the 0th row in the 9th depth
1D Arrays	
<code>a[:]</code>	Selects everything
<code>a[2:5]</code>	Selects the 2nd through the 4th rows (does not include the 5th row)
2D Arrays	
<code>b[:,:]</code>	Selects all rows and all columns
<code>b[:,0]</code>	Selects all rows, and the zeroth column
<code>b[0,:]</code>	Selects the zeroth row, and all columns in that row
<code>b[0:2,:]</code>	Selects the zeroth and first row, but NOT the second row
<code>b[0:2,0:2]</code>	Selects the zeroth and first row, and the zeroth and first column
3D Arrays	
<code>c[:, :, :]</code>	Selects all rows and columns on all depths
<code>c[0, :, :]</code>	Selects the everything in the first depth
<code>c[:, 0, :]</code>	Selects the first row of each depth
<code>c[:, :, 0]</code>	Selects the first column of each depth